

Deep Learning: Making Machines Ready for Complex Tasks

Deep learning or deep structured learning is a branch of machine learning based on a set of algorithms, which attempt to model high level abstractions in data by using multiple processing layers with complex structures. Deep learning is a science that's in the early stages of development

Deep learning is a new buzzword among machine learning researchers, both in academia and industry. It deals with making machines learn to solve complex real world tasks using approaches similar to the ones adopted by their human counterparts.

Computers are good at following instructions without even a single bit of deviation. The programs we write are simply a collection of instructions, which need to be executed in a pre-determined order. Is that good or bad? What these programs are good for is being determinable and precise with respect to their input and output mapping. At the same time, they are not so good while handling novel scenarios. The fundamental idea of machine learning is to make the machines analyse the underlying data model and develop their own 'understanding' so that they can react to

unforeseen scenarios. For example, certain tasks that are mundane for humans like recognising faces, voices, etc, don't fit into the realm of simply giving a predefined path to follow. They require the machines to 'learn' problem-solving instead of applying a readymade formula. For example, recognising a hand written number or digit is very different from finding out the area of a circle (for which a well-proven universal formula is available). This approach of making machines understand novel problems and provide a solution is called machine learning.

Machine learning is an umbrella term covering a spectrum of approaches. Some of them are listed below:

- Support vector machines
- Bayesian networks
- Artificial neural networks (ANN)
- Genetic algorithms
- Association rule learning, etc

Deep learning falls under the ANN category with certain specific features.

Artificial neural networks (ANN)

ANN is a method to train computers in a bio-inspired manner. You may be aware that the human brain is made up of super-speciality cells called neurons. These neurons are interconnected to form a neural network—the basic infrastructure for all the learning that we do. The uniqueness of this network is the size—the average human brain has 100 billion neurons and each neuron may have connections to 10,000 other neurons, thus making a gigantic network of 1000 trillion connections. Another interesting feature of these neural networks is that the human brain has evolved to its present form over thousands of years to achieve optimised results.

ANN is an attempt to mimic the biological brain using software instructions, which would give the computers the ability to learn things similar to the way we do. You might understand the mammoth nature of this task from the viewpoint of size and real-time responses. However, computer scientists have attempted ANN as early as the 1940s. The lack of advancements

